Rutgers-The State University
Master's Comprehensive Examination

December 4, 1982

Theory

Part I

Instructions:
1. Open book, and open notes
2. Do four out of the following six problems
3. Use a separate blue book for this section of the exam
4. Indicate which questions you wish graded

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1. Let $X_1,\ldots,X_n$ be independent Poisson random variables with $\lambda_i=E(X_i)$

   (i) Find the distribution of $Y=X_1+X_2+\ldots+X_n$

   (ii) Find the joint conditional distribution of $X_1,\ldots,X_n$ given Y.

2. Suppose $X_1,\ldots,X_n$ is a random sample from a $U(a,b)$ distribution, where both a and b are unknown.

   (i) What is a sufficient statistic for (a,b)?

   (ii) Find unbiased estimates of a and b based on the sufficient statistic.

3. Let $X_1,\ldots,X_n$ be a random sample from a distribution with density

   $f_\theta(x)=\theta\, x^{\theta-1}I_{(0,1)}(x)\quad \theta>0$

   Find a UMP level α test for H_0: $\theta\geq\theta_0$ vs H_1: $\theta<\theta_0$.

   For n=2 and $\theta_0=1$, find the cutoff point if $\alpha=.1$ and sketch the power function.

4. Suppose $X_1,\ldots,X_n$ are iid random variables whose moment generating function is always finite. Prove that if $\mu_1=E(X_1)$, $\mu_2=E(X_1^2)$ and $\mu_3=E(X_1^3)$

   then $E(\bar{X}^3)=\frac{1}{n^2}\mu_3+3\frac{n-1}{n^2}\mu_2\mu_1+\frac{(n-1)(n-2)}{n^2}(\mu_1)^3$

5. Let $X_1,\ldots,X_n$ be a random sample from the density $f_{\theta_1}(x)=\theta_1 x^{\theta_1-1}I_{(s,1)}(x)$

   and $Y_1,\ldots Y_m$ be a random sample from the density $f_{\theta_2}(y)=\theta_2 y^{\theta_2-1}I_{(0,1)}(y)$.

   (i) Find a generalized likelihood ratio test for testing H_0: $\mu_1=\mu_2$ vs H_1: $\mu_1\neq\mu_2$

   where μ_i is the mean of f_{θ_i}.

(ii) Let $Z_i = -\log X_i$ and $W = -\log Y_i$. Express the test derived in part (i) in terms of the statistic $T = \dfrac{\Sigma W_i}{\Sigma Z_i + \Sigma W_i}$

(iii) (optional) What is the distribution of T under H_0?

6. Let ρ be the unknown proportion of people in New Jersey who favor the nuclear freeze proposition. Two T.V. networks want to estimate p and their respective priors are:

$$\xi_1(p) = 2p \quad , 0 < p < 1$$

and

$$\xi_2(p) = 4p^3 \quad , 0 < p < 1 .$$

A sample of n N.J. residents is polled. Let X be the number of those polled who favored the proposition.

(i) Find the posterior distribution of p , for each of the two networks. (Hint: What type of densities are ξ_1, ξ_2?)

(ii) Find the Bayes estimate that each network would quote, if both use squared error loss.

(iii) Show that, no matter what value X has, the two Bayes estimates cannot differ by more than .002 , if n=1000.

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Applied

Part II

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1. The following contingency table describes the relation between scores above and below the median on an examination and ratings of job performance for 100 employees.

	Rating: Below average	Average	Above average	
Above median	11	25	35	71
Below median	15	7	7	29
	26	32	42	100

Test the hypothesis that job performance is independent of examination results. Is the test statistic you use legitimate in this situation, why, why not? If you rejected independence, what more can you say here?

2. Ten subjects were asked to perform a simple puzzle assembly task under normal conditions and under conditions of stress. During the stress condition, the subjects were told that a mild shock would be delivered 3 minutes after the start of the experiment and every 30 seconds thereafter until the task was completed. Blood pressure readings were taken under both conditions. The following data represent the highest reading under the experiment

Subject	Normal	Stress
1	126	130
2	117	118
3	115	125
4	118	120
5	118	121
6	128	125
7	125	130
8	120	120
9	127	125
10	119	123

Do the data present sufficient evidence to indicate higher reading during conditions of stress? Analyze the data using a nonparametric test. State the underlying assumptions. Why might not a t-test be valid in this situation? How would you find out if a t-test was valid?

Master's Comprehensive Examination

Part II
Applied

3. Suppose that each of two factors, A & B has been varied and that the results (responses Y's) are as follows

		B: 0	B: 1
A	0	66	82
	1	44	60

a) Summarize and describe the effects of each factor
b) What is an appropriate model for this situation?
c) Fit the model in (b)

If the results had been as below answer questions (a) above and (d)and (e) below.

		B: 0	B: 1
A	0	67	81
	1	43	61

d) What is an appropriate model if the value of σ (Y) is about 1.0? Fit the model.
e) What is an appropriate model if the value of σ(Y) is about 0.1 ? Fit the model.

4. It is well known that large bodies of water have a mitigating effect on the temperature of the surrounding land masses. On a cold night in central Florida, temperatures were recorded at equal distances along a transect running in the down-wind direction from a large lake. The data are as follows

Site (x)	1	2	3	4	5	6	7	8	8	9	10
Temperature (Y)	37.00	36.25	35.41	34.92	34.52	34.45	34.4	34.00	33.80	33.62	33.90

(a) Fit a linear model by least squares.
(b) In the light of question (a), if you could make only a single plot what would it be?
(c) Give a 99% confidence interval for a new observation taken at site 7.
(d) If there is any way to test for goodness of fit, do it, what do you conclude.

The following information may be helpful

$\Sigma X_i = 63 \qquad \Sigma X_i^2 = 449.00$

$\Sigma Y = 382.27 \qquad \Sigma Y_i^2 = 13296.084$

Part II
Applied

5. For all the planets of the solar system, distance (in astronomical units (AU)) from the sun is plotted against planet's period (in tropical years).
   (a) Examine the plot and describe what you see
   (b) Specify what you would like to see done to the plot to further clarify the relationship between distance and period.
   (c) How you might analyse the data after performing part (b) above, and what you would expect the computations to show
   (d) What further information would be helpful in understanding the relationship between period and distance.

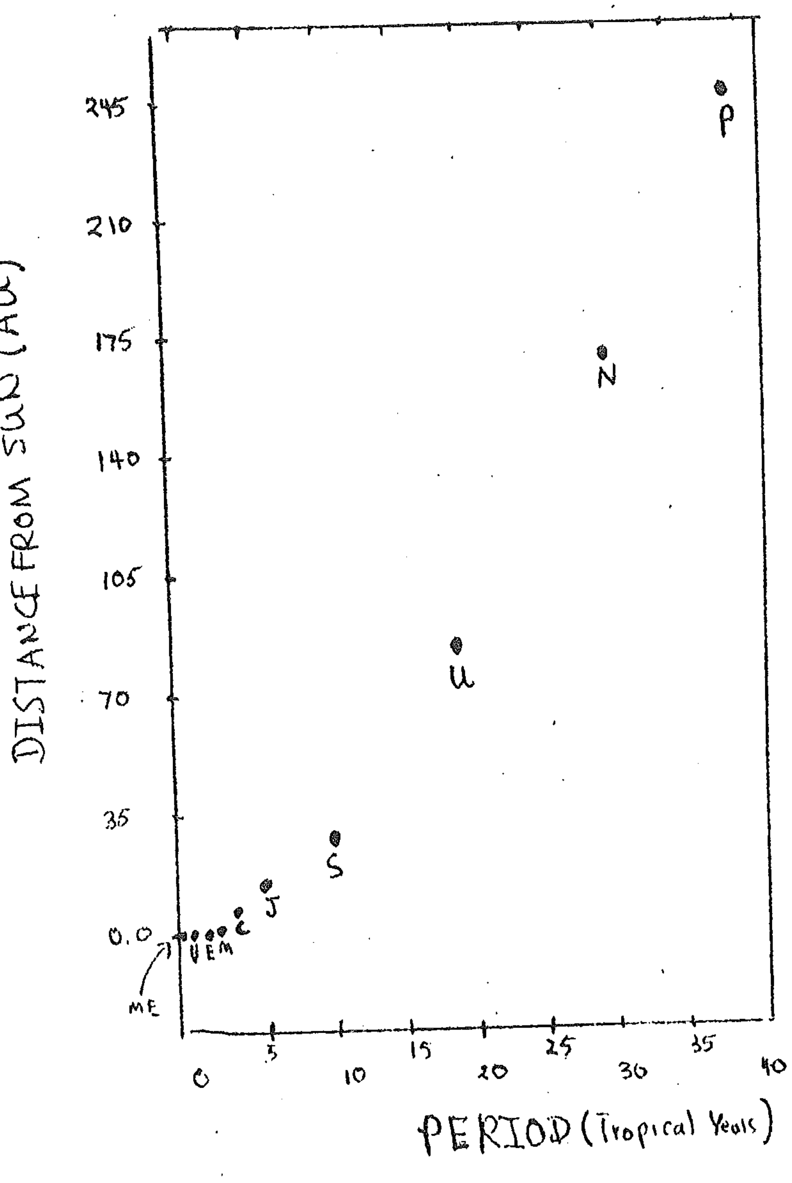
6. In an experiment infants were randomized to one of 3 treatments in an effort to determine if certain exercises would accelerate the time at which infants learned to walk by themselves. The results are below

Table: Ages of Infancts for Walking Alone (Months)

	active-exercise Group	passive-exercise Group	Control Group
	9.00	11.00	11.50
	9.50	10.00	15.00
	9.75	11.75	22.00
	9.60	11.8	21.00
Sum	37.85	44.55	69.50
Sum of Squares	358.47	498.30	1282.25

   (a) Construct an ANOVA Table for no differences at $\alpha=.05$
   (b) Test whether exercise as a whole makes a difference
   (c) Discuss briefly and succinctly anything that leads you to question the conclusions of the above analysis.

J Arnol
10/21/81



P = Pluto
N = Neptune
U = Uranus
S = Saturn
J = Jupiter
C = Ceres
M = Mars
E = Earth
V = Venus
ME = Mercury

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1. Let X_n denote the frequency of heads in n tosses of a fair coin. How many tosses do we need so that $P(.4 \le X_n \le .6)$ is at least .90?

2. A random variable X has moment generating function e^{t+18t^2}

   (a) Find the density of X.

   (b) Calculate $P(0 < X < 10)$.

3. Let $X_1, X_2, \ldots, X_n$ be i.i.d. Bernoulli r.v.'s with probability of success in a trial = p. Let δ be some unbiased estimator of $(1-p)^2$. Find a lower bound for the variance of δ.

4. A shooter hits a target 95% of the time. What is the expected number of times he hits the target before missing it for the first time?

5. A large shipment of books contains 2% books with imperfect binding. Use the Poisson approximation to binomial distribution to determine the probability that among 600 books (randomly selected from the shipment) at least 15 will have imperfect bindings.

6. Find a generalized likelihood ratio test of size α for testing $H_0: \theta \le 1$ versus $H_1: \theta > 1$ on the basis of a random sample $X_1, X_2, \ldots, X_n$

   from $f(x;\theta) = \theta\, e^{-\theta x}$, $x \ge 0$.

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APPLIED

Part II

Instructions:
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1. Consider the following random sample of size 40.

.18	.06	.27	.58	.98
.55	.24	.58	.97	.36
.48	.11	.59	.15	.53
.29	.46	.21	.39	.89
.34	.09	.64	.52	.64
.71	.56	.48	.44	.40
.80	.83	.02	.10	.51
.43	.14	.74	.75	.22

Test the hypothesis that this random sample came from a beta distribution with density

$$f(x) = 6x(1-x), \quad 0 < x < 1$$

(a) Divide the data into five classes. By doing a simple integration find the theoretical frequency corresponding to each class interval.

(b) Carry out the test for the hypothesis above at level $\alpha = .05$.

2. For many years sodium nitrite has been used as a curing agent for bacon but only recently it has been discovered that, during frying, the sodium nitrite induces the formation of nitrosopyrrolidine (NPy), a carcinogen. A study was done on four brands of bacon to determine the amount of NPy (in ppb) recovered after frying three slices from each brand. Here are the data:

Brand			
A	B	C	D
20	75	15	25
40	25	30	30
18	21	21	31

(a) Do the analysis of variance for this data set.

(b) Test the subhypothesis that the first two brands yield the same amount of NPy after frying. Do all tests at level $\alpha = .05$.

3. Heart rates were monitored for six laboratory animals during three different stages of their sleep: LSWS (light slow-wave sleep), DSWS (deep slow-wave sleep) and REM (rapid eye-movement sleep). Here are the results: (heart rate = # of beats/5 seconds)

Animal	LSWS	DSWS	REM
1	14.1	11.7	15.7
2	26.0	21.1	21.5
3	20.9	19.7	18.3
4	19.0	18.2	17.0
5	26.1	23.2	22.5
6	20.5	20.7	18.9

(a) Do the analysis of variance to test the equality of the heart rates during the three phases of sleep. Let $\alpha = .05$.

(b) There is a marked physiological difference between REM sleep and LSWS and DSWS sleep. Test the REM rate against the average of the other two. Is your test significant at the .05 level?

4. Write down a $2^5/4$ fractional replicate design with factors A, B, C, D and E, taking ABC and CDE as defining contrasts. What are the alias groups? Take some hypothetical data for yield and prepare the ANOVA table for this design assuming that all the interactions are negligible.

5. The following are the number of misprints counted on pages selected at random from three Sunday New York Times:

Sunday 1: 4, 10, 2, 6, 4, 12

Sunday 2: 8, 5, 13, 8, 8, 10

Sunday 3: 7, 9, 11, 2, 14, 7

Use the Kruskal-Wallis test at the level of significance $\alpha = .05$ to test the null hypothesis that the three samples come from identical populations, against the alternative that the compositors or the proofreaders who worked on the three Sunday editions were not equally good.

6. The following data were collected on emphysema patients: X represents the number of years the patient smoked and Y represents an evaluation of the extent of lung damage (on a scale of 1 to 100).

X	25	36	22	15	48	39	42	31	28	33
Y	55	60	50	30	75	70	70	55	30	35

(i) Find the least squares regression line of Y on X.

(ii) Test the hypothesis that the correlation coefficient between X,Y is 0. Use $\alpha = .05$.

(iii) Find a 95% confidence interval for the slope of the regression line you computed in part (i). Would you reject the hypothesis that the slope is zero at level $\alpha = .01$?